

Firas Selmi

Network Engineer | Network Automation

Allendale, MI | Phone: 616-313-5037 | Email: firass@mail.gvsu.edu
linkedin.com/in/selmi-firas | github.com/Selmi-Firas

SUMMARY

Junior network engineer with hands-on experience deploying and troubleshooting LAN and wireless infrastructure, switches, access points, and firewalls. designing segmented, access-controlled network architectures. Automation foundation in Ansible and Python, with working knowledge of Linux, virtualization, and industrial protocols. M.S. in Cybersecurity, Grand Valley State University, GPA 3.4, 2026; CCNA in progress

TECHNICAL SKILLS

Networking: Cisco Routing & Switching, VLANs, Access Switches, Wireless Access Points, TCP/IP, Ethernet, DHCP, DNS, NAT, QoS, Structured Cabling, Cable Testing & Port Mapping
Security: 802.1X / Network Access Control, FreeRADIUS, Network Segmentation, Zero Trust, Firewalls (pfSense, Cyberoam), Multi-WAN, Content Filtering, IDS/IPS (Suricata, Snort), SIEM/SOC (Wazuh, ELK)
Automation & AI: Ansible, Python, Bash, Configuration Deployment, Network Automation, LLM Integration, langgraph
Industrial & Embedded: CAN Bus (MCP2515), Serial/SPI, Sensor Telemetry, Protocol Translation, Raspberry Pi, Arduino
Systems & Tools: Linux (Ubuntu, CentOS), Windows Server, Proxmox, Docker, Wireshark, Nmap, GNS3, Cisco Packet Tracer

EXPERIENCE

Systems & Network Administrator (Graduate Assistant) Grand Valley State University 09/2025 – 05/2026
Systems & network support for classroom labs and virtualization infrastructure Allendale, MI

- Built a six-playbook Ansible automation suite managing 31 Windows workstations over WinRM silent unattended software deployment with idempotency checks, bulk file distribution, and network-adaptor discovery reducing lab-wide provisioning from per-machine manual work to a single command.
- Automated full VM lifecycle deployment across all workstations: OVA transfer, headless VirtualBox import, bridged-network configuration, and per-machine credential injection via VBoxManage guest control.
- Administered a Proxmox VE server (32 TB storage, 128 GB RAM, 16-core) hosting the lab's virtualization and Ansible control infrastructure, and supported faculty across concurrent courses on a 31-workstation network.

Network & Systems Administrator Placeholder Infrastructures Groupe EVACOM 01/2024 – 08/2024
Network architecture, security tooling, and team leadership Tunisia

- Designed and led implementation of a zero-trust network architecture, segmenting IT and OT domains behind dedicated firewalls with an isolated DMZ, and building 802.1X/FreeRADIUS network access control across wireless infrastructure.
- Installed, configured, and maintained LAN and wireless equipment 2 Cisco access switches, 3 wireless access points, and a pfSense firewall provisioning VLANs, DHCP, DNS, and NAT for 24 users and 20+ devices.
- Deployed switches and servers across 35 workstations and 4 servers, pulling and testing cable runs, tracing links across floors with cable testers, and mapping switch ports for 30–40 users.
- Directed 7 interns across security infrastructure workstreams; responded to network incidents using monitoring tooling (Wazuh, Suricata, Snort) and authored network documentation and standard procedures.

Network Technician Placeholder Infrastructures Groupe EVACOM 06/2023 – 12/2023
IT support and Proxmox environment maintenance Tunisia

- Resolved connectivity, hardware, and access issues for 30–40 users and operated virtualized server infrastructure, handling access provisioning and weekly backups.

IT Intern Placeholder Infrastructures Groupe EVACOM 01/2023 – 05/2023
Server deployment and IDS/IPS monitoring Tunisia

- Built an isolated test network on a Dell PowerEdge R360 (Proxmox VE, PERC RAID 10), deploying Security Onion (Suricata, Snort, Zeek, Wazuh, ELK) as a standalone IDS platform and a Windows Server 2016 domain controller (Active Directory, DNS, DHCP, FTP, IIS); validated monitoring with a simulated DoS attack and confirmed detection.

PROJECTS

Zero-Trust Network Architecture Designed a segmented IT/OT enterprise network (topology, IP addressing scheme, security zones, DMZ) replacing a flat, unsegmented network, routing each domain into separate Layer 3 networks to create enforceable inter-zone boundaries; deployed dual firewalls (pfSense, Cyberoam) and implemented 802.1X/FreeRADIUS network access control (PEAP) across the wireless infrastructure. Validated on an isolated replica with confirmed IDS/IPS detection.

SDN Control Plane for a Distributed Network (M.S. Thesis) Designed a centralized SDN-style control plane over a 17-node network: active telemetry, threshold-based alerting, quality-ranked path selection, EDF scheduling, and closed-loop reliability feedback. Measured a 6.6-point gain in task success over a state-blind baseline across 10,000 runs (Python; simulated quantum network).

EDUCATION

M.S. Cybersecurity 05/2026
Grand Valley State University Allendale, MI

B.Eng. Network and Systems Engineering 05/2023
Higher Institute of Applied Sciences and Technology of Mateur Tunisia

CERTIFICATIONS & LANGUAGES

Certification: CCNA (Cisco Certified Network Associate) In Progress, Expected 08/2026

Languages: Arabic (Native), French (Professional), English (Fluent)